The following B.E.S.T. Standards will be covered in this unit:

MA.912.F.1.1 Given an equation or graph that defines a function, determine the function type. Given an input-output table, determine a function type that could represent it.

- *Clarification 1: Within the Algebra 1 course, functions represented as tables are limited to linear, quadratic, and exponential.*
- *Clarification 2: Within the Algebra 1 course, functions represented as equations or graphs are limited to vertical or horizontal translations, or reflections over the x-axis of the following parent functions: $f(x) = x, f(x) = x^2, f(x) = x^3$, $f(x) = \sqrt{x}, f(x) = \sqrt[3]{x},\ f(x) = |x|, f(x) = 2^x$ and $f(x) = \left(\frac{1}{2}\right)^x$.*

MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.

- *Clarification 1: Problems include simple functions in two variables such as $f(x, y) = 3x - 2y$.*
- *Clarification 2: Within the Algebra 1 course, functions are limited to one-variable such as $f(x) = 3x$.*

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

- *Clarification 1: Instruction includes the use of technology when appropriate.*
- *Clarification 2: Refer to the K-12 Formulas (Appendix E) for the Laws of Exponents.*
- *Clarification 3: Instruction includes converting between expressions involving rational exponent and expressions involving radicals.*
- *Clarification 4: Within the Mathematics for Data and Financial Literacy course, it is not the expectation to generate equivalent numerical expressions.*

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.

- *Clarification 1: Within the Algebra 1 course, expressions are limited to a single arithmetic operation involving two square roots or two cube roots.*

Unit 1, Lesson 1: Classifying Parent Functions

Benchmark

MA.912.F.1.1 Given an equation or graph that defines a function, classify the function type. Given an input-output table, determine a function type that could represent it.

Learning Target

☐ I can classify families of functions from a graph or equation.

1.1.1 Warm-Up: Would You Rather?

1. Complete the chart by moving around the room and asking nine different people "Would you rather ...?" Introduce yourself to each person. Ask each person a different question from your chart. Record their name and their answer.

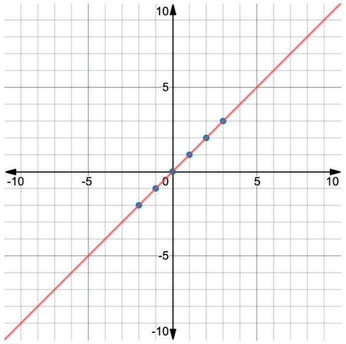
Would you rather ...

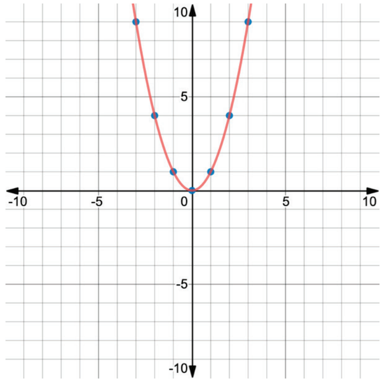
fly a kite or swing on a swing?	be a famous inventor or a famous writer?	go skiing or go to a water park?
be able to create a new holiday or create a new sport?	start a colony on another planet or be the leader of a small country on Earth?	own an old pirate ship with crew or a private jet with a pilot?
see a firework display or a circus performance?	be able to talk to land animals or talk to sea animals?	time-travel to the past or the future?

1.1.2 Activity: Parent Functions Gallery Walk

The study of functions is an area of focus in algebra. Functions are classified into families. The ***parent function*** is the simplest function of each family that preserves the characteristics of the family.

Here are eight parent functions that will be introduced in Algebra 1. After you examine each parent function, answer the questions that follow.

<table>
<tr><th colspan="3">Linear Function</th></tr>
<tr><th>Parent Function</th><th>Table</th><th>Graph</th></tr>
<tr><td>$y = x$</td><td>
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-2</td><td>-2</td></tr>
<tr><td>-1</td><td>-1</td></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td></tr>
<tr><td>2</td><td>2</td></tr>
<tr><td>3</td><td>3</td></tr>
</table>
</td><td>
</td></tr>
<tr><td>What do you notice about the graph?</td><td colspan="2"></td></tr>
<tr><td>What do you notice about the equation?</td><td colspan="2"></td></tr>
</table>

<table>
<tr><th colspan="3">Quadratic Function</th></tr>
<tr><th>Parent Function</th><th>Table</th><th>Graph</th></tr>
<tr><td>$y = x^2$</td><td>
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-2</td><td>4</td></tr>
<tr><td>-1</td><td>1</td></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td></tr>
<tr><td>2</td><td>4</td></tr>
<tr><td>3</td><td>9</td></tr>
</table>
</td><td>
</td></tr>
<tr><td>What do you notice about the graph?</td><td colspan="2"></td></tr>
<tr><td>What do you notice about the equation?</td><td colspan="2"></td></tr>
</table>

Cubic Function

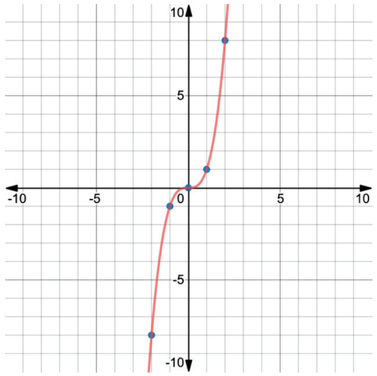
Parent Function

$y = x^3$

Table

x	y
-2	-8
-1	-1
0	0
1	1
2	8
3	27

Graph



What do you notice about the **graph**?	
What do you notice about the **equation**?	

Square Root Function

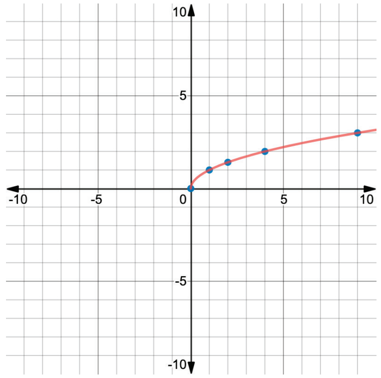
Parent Function

$y = \sqrt{x}$

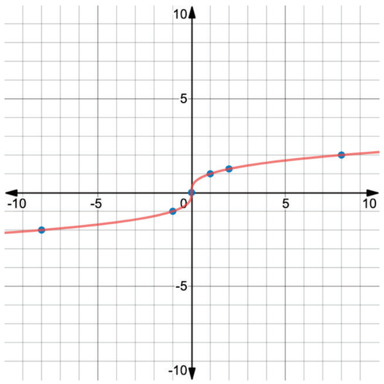
Table

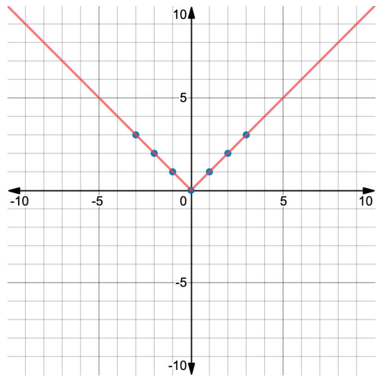
x	y
-1	not real
0	0
1	1
2	1.414
4	2
9	3

Graph

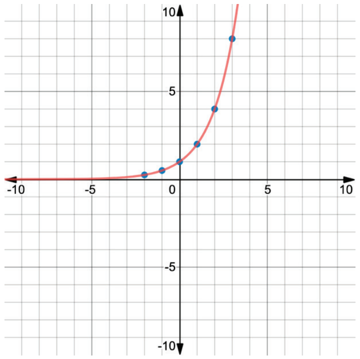


What do you notice about the **graph**?	
What do you notice about the **equation**?	

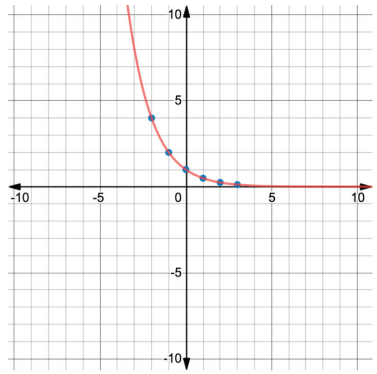
<table>
<tr><th colspan="3">Cube Root Function</th></tr>
<tr><th>Parent Function</th><th>Table</th><th>Graph</th></tr>
<tr><td>$y = \sqrt[3]{x}$</td><td>
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-8</td><td>-2</td></tr>
<tr><td>-1</td><td>-1</td></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td></tr>
<tr><td>2</td><td>1.259</td></tr>
<tr><td>8</td><td>2</td></tr>
</table>
</td><td></td></tr>
<tr><td>What do you notice about the graph?</td><td colspan="2"></td></tr>
<tr><td>What do you notice about the equation?</td><td colspan="2"></td></tr>
</table>

<table>
<tr><th colspan="3">Absolute Value Function</th></tr>
<tr><th>Parent Function</th><th>Table</th><th>Graph</th></tr>
<tr><td>$y = |x|$</td><td>
<table>
<tr><th>x</th><th>y</th></tr>
<tr><td>-2</td><td>2</td></tr>
<tr><td>-1</td><td>1</td></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td></tr>
<tr><td>2</td><td>2</td></tr>
<tr><td>3</td><td>3</td></tr>
</table>
</td><td></td></tr>
<tr><td>What do you notice about the graph?</td><td colspan="2"></td></tr>
<tr><td>What do you notice about the equation?</td><td colspan="2"></td></tr>
</table>

<table>
<tr><th colspan="3">Exponential Function</th></tr>
<tr><th>Parent Function</th><th>Table</th><th>Graph</th></tr>
<tr><td>Exponential Growth Base (2)

$y = 2^x$</td><td><table><tr><th>x</th><th>y</th></tr><tr><td>-2</td><td>$\frac{1}{4}$</td></tr><tr><td>-1</td><td>$\frac{1}{2}$</td></tr><tr><td>0</td><td>1</td></tr><tr><td>1</td><td>2</td></tr><tr><td>2</td><td>4</td></tr><tr><td>3</td><td>8</td></tr></table></td><td>
</td></tr>
<tr><td>What do you notice about the graph?</td><td colspan="2"></td></tr>
<tr><td>What do you notice about the equation?</td><td colspan="2"></td></tr>
</table>

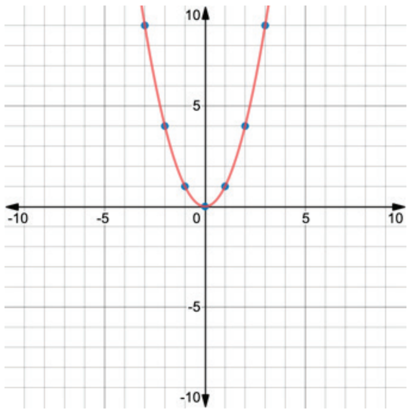
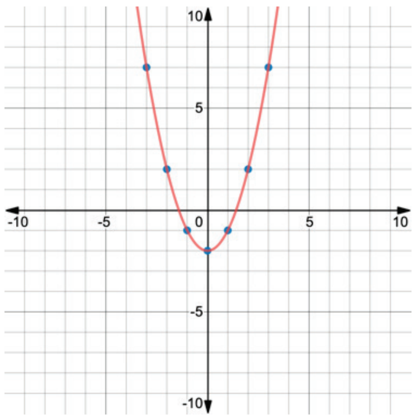
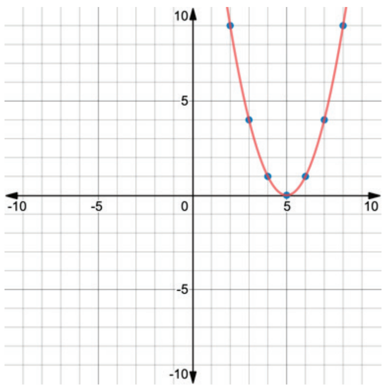
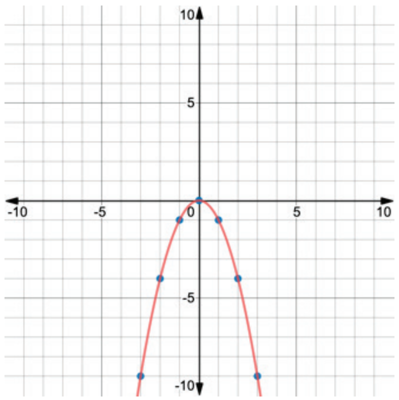
<table>
<tr><th colspan="3">Exponential Function</th></tr>
<tr><th>Parent Function</th><th>Table</th><th>Graph</th></tr>
<tr><td>Exponential Decay Base $\left(\frac{1}{2}\right)$

$y = \left(\frac{1}{2}\right)^x$</td><td><table><tr><th>$x$</th><th>$y$</th></tr><tr><td>$-2$</td><td>$4$</td></tr><tr><td>$-1$</td><td>$2$</td></tr><tr><td>$0$</td><td>$1$</td></tr><tr><td>$1$</td><td>$\frac{1}{2}$</td></tr><tr><td>$2$</td><td>$\frac{1}{4}$</td></tr><tr><td>$3$</td><td>$\frac{1}{8}$</td></tr></table></td><td>
</td></tr>
<tr><td>What do you notice about the graph?</td><td colspan="2"></td></tr>
<tr><td>What do you notice about the equation?</td><td colspan="2"></td></tr>
</table>

1.1.3 Guided Practice: Classifying Functions

Functions can be transformed by translations, dilations, and reflections.

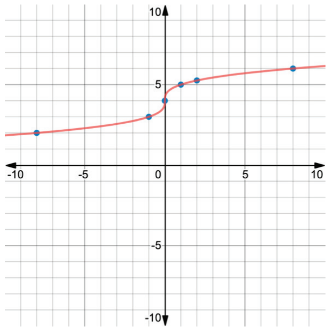
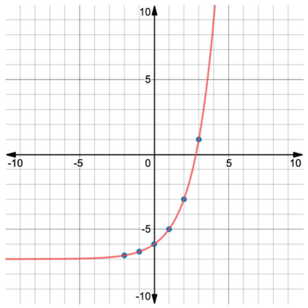
In this example, the quadratic parent function is translated vertically, horizontally, and reflected over the x-axis.

Quadratic Parent Function	Translated Vertically
$y = x^2$	$y = x^2 - 2$
	
Translated Horizontally	**Reflected over the x-axis**
$y = (x-5)^2$	$y = -(x)^2$
	

1. What stayed the same in each of the equations?

2. What stayed the same in each of the graphs?

3. Classify each function into its function family based on the characteristics observed for the parent functions.

Function	Function Family
$y = \lvert x + 9 \rvert$	
$y = -\left(\frac{1}{2}\right)^x$	
$y = x^3 - 6$	
	
	

1.1.4 Your Turn!

Classify the function family for each equation or graph.

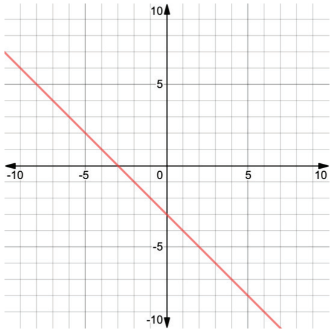
1. $y = (x+2)^3$ Function Family:________________

2. $y = -x - 23$ Function Family:________________

3. $y = 2^{(x-7)}$ Function Family:________________

4. $y = (x)^2 + 31$ Function Family:________________

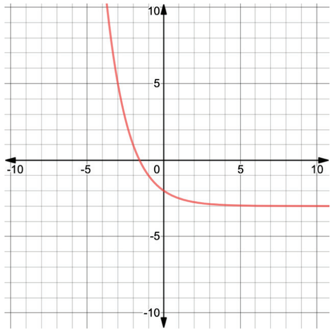
5.



Function Family:

____

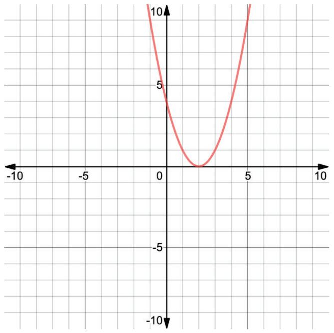
6.



Function Family:

____

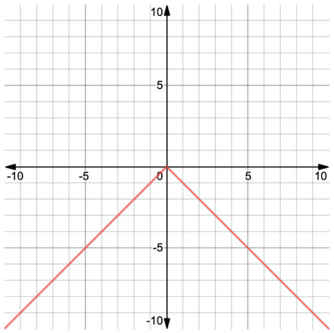
7.



Function Family:

____

8.



Function Family:

____

1.1.5 Wrap-Up: Reflection

1. As you begin your journey through Algebra 1, rate yourself on where you feel you are today on the following skills.

Skill	**1 – I am just starting.** **2 – I have some skills.** **3 – I am almost there.** **4 – I have it!**
I can evaluate expressions using order of operations.	
I can graph points on a coordinate plane.	
I can apply the laws of exponents.	
I can determine whether a relation is a function.	

2. Rate yourself based on how you feel about the following statements.

Statement	**1 – I struggle with this.** **2 – I am ready to grow in this area.** **3 – I am almost there.** **4 – That's me!**
I am willing to try new things.	
I persevere when something is challenging.	
I am a student learner.	
I am a mathematician.	

Unit 1, Lesson 2: Evaluating Functions in Function Notation

Benchmark

MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.

Learning Target

☐ I can evaluate a function for a specific input value of the domain.

1.2.1 Warm-Up: Food and Research Career

Richard works in Research and Development for Kellogg's. One of the projects he is most proud of involves developing a line of organics products and no added sugar products. Richard also points out that working with a team provides an exciting variety of experiences that is crucial and rewarding to the work of the company.

1. How could mathematics help in the food science industry?

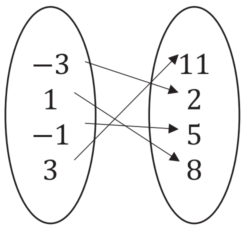
2. Why is it important to have teamwork skills when working for a company?

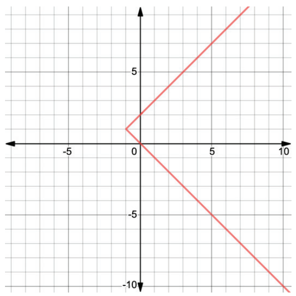
1.2.2 Collaboration: Functions

A ***relation*** is a set of input-output pairs.

A ***function*** is a mathematical relation for which each element in the domain corresponds to exactly one element in the range.

Three different relations are shown, with each represented in a different way.





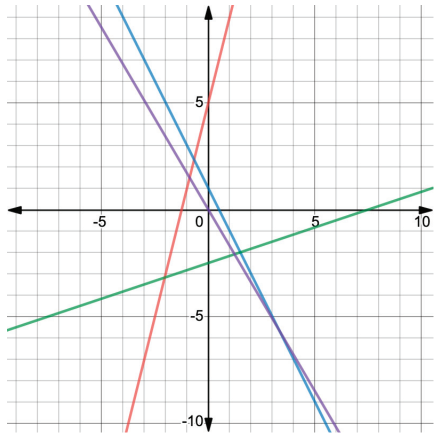
x	y
-4	2
7	0
-4	-4
0	-6

1. With your partner, write an explanation identifying which of the relations above are functions.

1.2.3 Exploration: Graphs of Functions

The equations and graphs of four linear functions are shown.

$y = 4x + 5$
$y = -2x + 1$
$y = \frac{1}{3}x - 2.5$
$y = -1.7x$



1. Write the equations of the lines that decrease from left to right.

2. Write the equations of the lines that increase from left to right.

3. What do the equations of the lines that increase from left to right have in common?

4. How are the equations of the lines that decrease from left to right different from the equations of the lines that increase from left to right?

1.2.4 Guided Instruction: Evaluating Functions

Each of the functions in the Exploration starts with "$y =$." If asked to use the function y from the list of functions to find a specific value, it would not be clear which of the four functions to use.

Since mathematicians often work with multiple functions at a time, ***function notation*** is used to name specific functions.

The notation $y = f(x)$ defines a function named f. This indicates "f is a function of x." The letter x represents the input value, or independent variable. The letter y, or $f(x)$, represents the output value, or dependent variable. The notation $f(x)$ is read, "f of x."

name of function   input of function

$$\underbrace{f(x)}_{\text{output of function}}$$

A function can be named using any letter. For example, the previous functions can be renamed as shown:

$f(x) = 4x + 5$ $g(x) = -2x + 1$

$h(x) = \frac{1}{3}x - 2.5$ $c(x) = -1.7x$

Now, if asked to use the function h to find a specified value of x, it would be clear which function to use.

Functions are evaluated in the same way equations are evaluated when given a value for a variable.

Consider the function $f(w) = -7w^2 - 10w + 3$.

1. Find the value $f(10)$.

$$f(w) = -7w^2 - 10w + 3$$

$$f(10) = -7(10)^2 - 10(10) + 3$$

$f(10) =$ ________

When the input is 10, the output of the function is _____.

Using function notation, this is written $f(10) =$ _____.

2. Find the value $f(-3)$.

$f(-3) =$ _______

This same function $f(w)$ can be represented in other ways.

Shown here are a table of values, a graph, and a mapping diagram. The function can be evaluated from each of these representations.

Table of Values

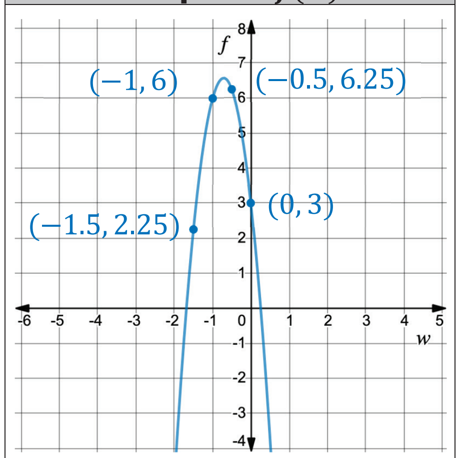
w	$f(w)$
-15	-1422
-10	-597
-1	6
2	-45

3. Find $f(-10)$.

   $f(-10) =$ _____

4. If $f(w) = -45$, what is the value of w?

Graph of $f(w)$

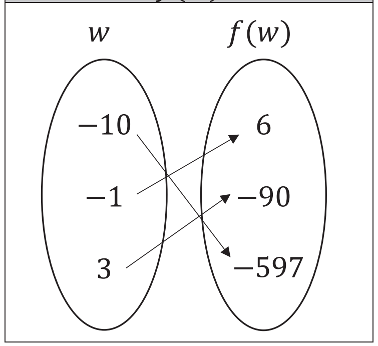


Find each value from the graph.

5. $f(-1.5) =$ _____

6. $f(0) =$ _____

Mapping Diagram of $f(w)$



Find each value from the mapping diagram.

7. $f(3) =$ _____

8. Find the value of w when $f(w) = 6$.

1.2.5 Try It: Function Notation

1. Given the function $c(y) = -6y - 5$, find each value.

   a. $c(-4) =$

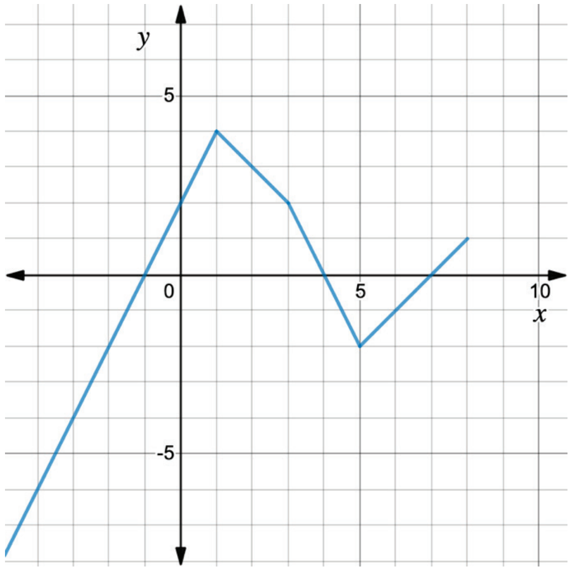
   b. $c\left(\frac{3}{4}\right) =$

   c. For what value of y is $c(y) = 9$?

2. Given $g(k) = 2k^2 - 4$, find $g(-5)$.

3. Given $h(x) = \sqrt{2.5x + 1}$, find $h(3.2)$.

4. Use the graph of $y = f(x)$ below to find each value.



   a. $f(1) =$

   b. $f(-2) =$

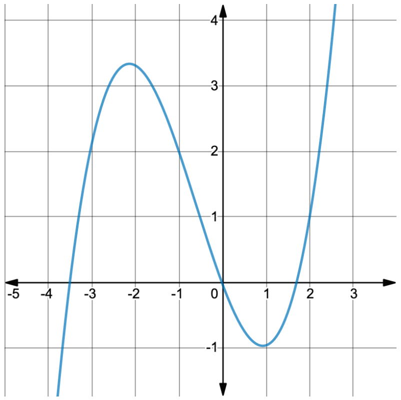
   c. For what value of x is $f(x) = -4$?

1.2.6 Your Turn!

1. Find each value given the function $m(t) = \frac{t}{3} + 6$.
   a. $m(-9) =$

   b. Find the input for which $m(t) = 13$.

2. Find each value using the graph of the function $f(x)$.



   a. $f(-3.5) =$

   b. $f(2) =$

   c. Find x when $f(x) = 4$.

3. Use the table of values representing $D(r)$ to find each value.

r	$D(r)$
-6	9
0	-5
-4	9
2	0
9	7

   a. $D(-4) =$

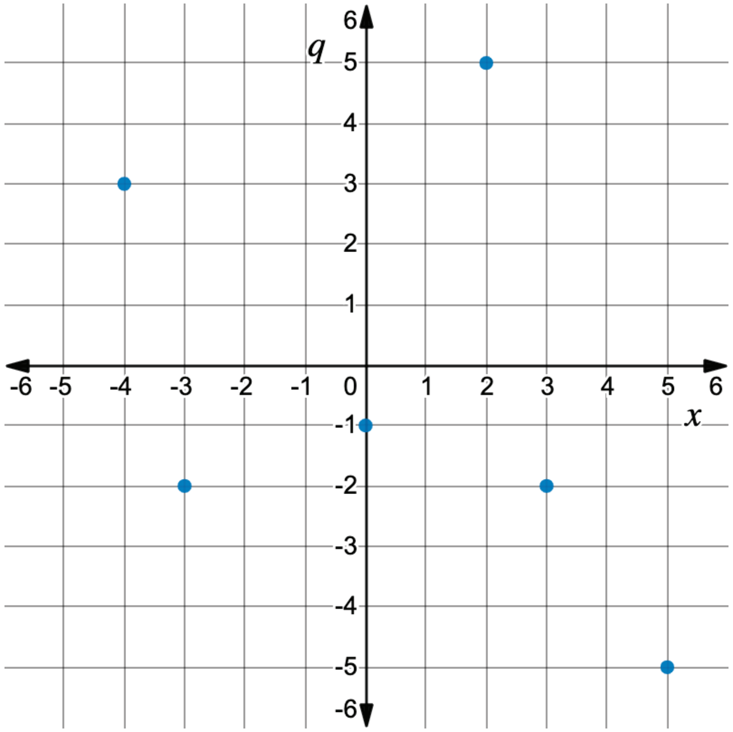
   b. $D(-6) =$

   c. Is $D(r)$ a function? Explain why or why not.

1.2.7 Wrap-Up: Ticket Out the Door

1. Given the function $h(t) = t^3 - 2t$, find $h(-7)$.

2. Use the graph of $q(x)$ to find $q(-3)$.



Unit 1, Lesson 3: Interpreting Function Notation in the Real World

Benchmark

MA.912.F.1.2 Given a function represented in function notation, evaluate the function for an input in its domain. For a real-world context, interpret the output.

Learning Target

- [ ] I can evaluate a function for a specific input value of the domain and interpret it in a real-world context.

1.3.1 Warm-Up: Equivalent Function Values

1. Fill in the boxes to construct two functions where the final equality statement is true. Use the digits 1 to 9 once each at most.

$$g(x) = \square x - \square$$

$$g(\square) = 26$$

1.3.2 Guided Instruction: Interpreting Functions

Recall that functions have a special notation which allows unique relationships to be expressed in a concise and meaningful way.

The value of a function is dependent on the input. The statement $D(2) = 60$ means that for an input of 2, the value of the function, D, is 60.

Functions can be used to model real-world relationships.

1. If $F(t)$ models the outside temperature in degrees Fahrenheit t hours after 6 a.m., the statement $F(2) = 60$ has a specific meaning within the context.

   $F(2) = 60$ can be interpreted as ______ hours after ____ a.m., which is ______ a.m., the outside temperature is ______ degrees Fahrenheit.

2. Physics students drop a ball from the top of a 400 foot high building and model its height above the ground as a function of time using the function $H(t) = 400 - 16t^2$. The height, H, is measured in feet and time, t, is measured in seconds.

   a. Find the value of $H(0)$.

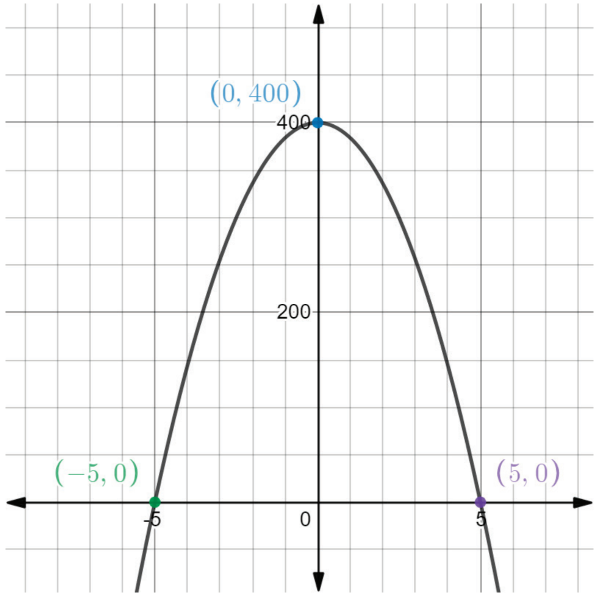
   b. What does $H(0)$ mean in the context of this problem?

   c. Evaluate $H(2)$.

   d. Interpret the meaning of $H(2)$ based on the given context.

   e. José is asked to interpret the value of $H(-3)$. Is interpreting this value reasonable in the context of this situation?

3. The graph of $H(t)$ is shown on the coordinate grid.



a. Based on the context, the x-axis represents ________________ and the y-axis represents _________________.

b. What will the height of the ball be when it hits the ground?

c. Circle the point where the ball will hit the ground.

d. Write the point where the ball hits the ground using function notation.

$$H(\_\_\_\_) = \_\_\_\_\_\_$$

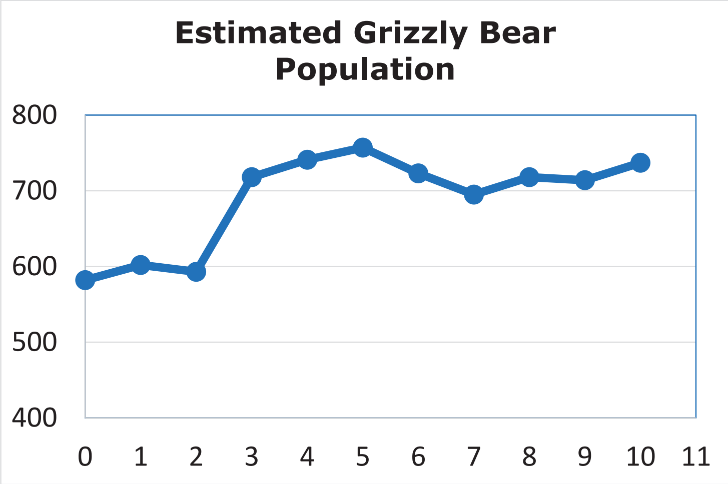
e. How long did it take the ball to hit the ground after it was released?

f. Discuss with your partner what $H(1.8) > H(3.4)$ means in terms of the context.

1.3.3 Guided Practice: Interpreting Functions

1. The population of grizzly bears at Yellowstone National Park has been threatened with local extinction since 1975. Through many conservation efforts, grizzlies have made a remarkable recovery. The function $y = G(t)$ is modeled below, where t is the number of years since 2009 and G is the number of grizzly bears.

   a. Label the x and y axes based on the context.



   b. Find the approximate value from the graph above.

   $$G(3) \approx \_\_\_\_\_\_$$

   c. In the context of this problem, $G(3)$ means that 3 years after 2009, there were approximately _______ bears in Yellowstone.

   d. Find the approximate value from the graph.

   $$G(8\tfrac{1}{2}) \approx \_\_\_\_\_\_$$

   e. What does $G(8\tfrac{1}{2})$ mean in the context of this problem?

   f. Explain why there is not a value of t, such that $G(t) = 710.5$.

2. Matthew filled his bathtub, took a bath, and then drained the tub. The function $B(m)$ gives the depth of the water, in inches, m minutes after he began to fill the bathtub.

   a. Explain the meaning of each statement in this context.

   $B(0) = 0$

   $B(9.5) = B(20)$

   b. Discuss with your partner why the inequality $B(1) < B(7)$ is likely true in this context.

3. A biologist is studying the relationship between a tree's diameter and its height. She records the following data for seven different trees using the function $H(d)$, where H represents the height, in feet, and d represents the diameter, in inches.

Diameter (inches)	2	4	6	8	10	12	14
Height (feet)	6	7	10	15	18	22	28

   a. Fill in the blanks in the following sentence to interpret the statement $H(10) = 18$ in this context.

   A tree with a diameter of ______ has a height of ______.

   b. Determine the value of $H(14)$.

   c. Interpret $H(14)$ for the given context.

1.3.4 Try It: Evaluating and Interpreting Functions

1. For groups of 80 or more people, a charter bus company determines the rate per person according to the formula $R(n) = 8 - 0.05(n - 80)$, where n represents the number of people and R represents the rate, in dollars, charged to each customer.

   a. Find each value and interpret it in context.

   $R(75) =$ _______

   $R(85) =$ _______

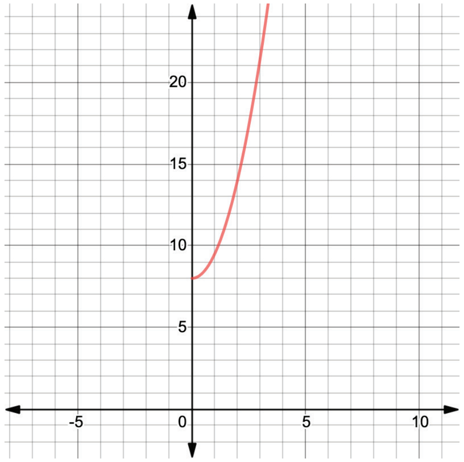
   b. Why is $R(75) > R(85)$ in this context?

1.3.5 Your Turn!

1. The size of a fungal colony is modeled by the equation $z(t) = \frac{3}{2}t^2 + 8$, where z is the size of the colony in microns, and t is the time, in hours.

   A graph of the function $z(t)$ is shown.

   Using the graph, approximate the values and interpret their meaning in this context.



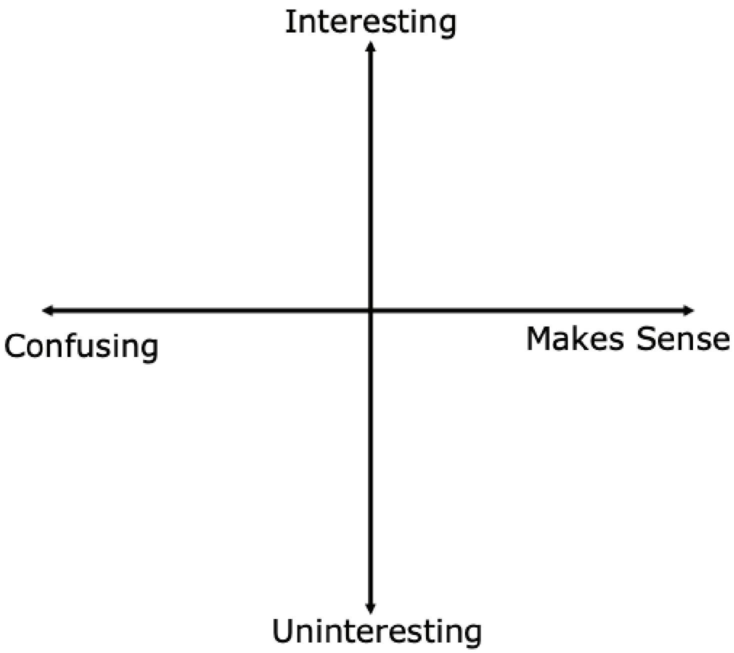
   a. $z(3) =$ _______

   b. $z(0) =$ _______

   c. Why does the graph only display y-values for the function when $t \geq 0$?

1.3.6 Wrap-Up: Reflection

1. Plot a point in one of the quadrants below to indicate where you feel you are on today's learning target: "I can evaluate a function for a specific input value of the domain and interpret it in a real-world context."



Unit 1, Lesson 4: Applying the Laws of Exponents

Benchmark

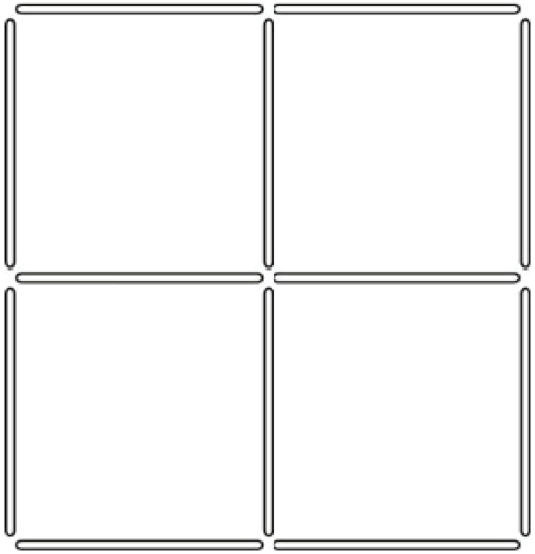
MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Learning Targets

- ☐ I can apply the Laws of Exponents to write equivalent numerical expressions.
- ☐ I can evaluate square roots and cube roots.

1.4.1 Warm-Up: Toothpick Squares

1. Make two squares of different sizes by removing two toothpicks.



1.4.2 Refresher: Laws of Exponents

1. Using the given exponent law, complete the table by writing an equivalent exponential expression.

Exponent Law	Expression	Equivalent Expression
Product of Powers $a^m \cdot a^n = a^{m+n}$	$6^5 \cdot 6^2$	
Quotient of Powers $\frac{a^m}{a^n} = a^{m-n}$	$\frac{5^{14}}{5^2}$	
Power of a Power $(a^m)^n = a^{m \cdot n}$	$(4^{10})^3$	
Power of a Product $(ab)^m = a^m \cdot b^m$	$(5 \cdot 2)^4$	
Power of a Quotient $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$	$\left(\frac{3}{11}\right)^2$	
Negative Exponents $a^{-1} = \frac{1}{a}$ $\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$ $\left(\frac{a}{b}\right)^{-m} = \frac{b^m}{a^m}$	9^{-1} $\left(\frac{5}{7}\right)^{-1}$ $\left(\frac{13}{4}\right)^{-3}$	
Identity Exponent $a^1 = a$	572^1	
Zero Exponent $a^0 = 1$	572^0	

1.4.3 Guided Instruction: Building on Square Roots and Cube Roots

To square a number means to

- ○ multiply it by itself.
- ○ multiply it by 2.

$$7^2 = \_\_\_\_ \cdot \_\_\_\_ = \_\_\_\_$$

$$(-7)^2 = \_\_\_\_ \cdot \_\_\_\_ = \_\_\_\_$$

The inverse of squaring a number is taking the square root ($\sqrt{}$).

index $\sqrt[n]{x}$ radicand

To evaluate $\sqrt[2]{25}$, also written as $\sqrt{25}$, ask, "What times itself equals 25?"

$$5 \cdot 5 = 25 \text{ or } -5 \cdot -5 = 25$$

Therefore, $\sqrt{25}$ is 5 and -5.

All radicands greater than or equal to 0, $x \geq 0$, have two square roots, a positive and a negative. The positive square root is referred to as the ***principal square root***.

The square root of 16, written as $\sqrt{16}$, is 4 and -4 because both 4^2 and $(-4)^2$ are equal to 16.

To cube a number means to

- ○ multiply the number by 3.
- ○ triple the number.
- ○ multiply the number by itself three times.

$6^3 =$ $\qquad\qquad$ $(-6)^3 =$

The cube root of a number is a value that when multiplied by itself three times gives that number.

$$27 = \_\_\_\_^3 \text{ so } \sqrt[3]{27} = \_\_\_\_$$

$$-27 = \_\_\_\_^3 \text{ so } \sqrt[3]{-27} = \_\_\_\_$$

The cube root of 64, written as $\sqrt[3]{64}$, is 4 because 4^3 is 64.

$$4 \cdot 4 \cdot 4 = 64$$

1.4.4 Your Turn!

1. Find the value(s) of each.

   a. $\sqrt{121} =$ ______

   b. $\sqrt[3]{125} =$ ______

2. Work with a partner to complete the chart.

Expression	Exponent Law	Equivalent Expression
$8^2 \cdot 8^4$	Product of Powers	
	Product of Powers	10^7
$(7^5)^3$	Power of a Power	
	Power of a Power	4^{32}
$\frac{2^{12}}{2^{10}}$	Quotient of Powers	
	Quotient of Powers	6^9
$(3 \cdot 7)^5$	Power of a Product	
	Power of a Product	$13^5 \cdot 2^5$
$\left(\frac{1}{7}\right)^4$	Power of a Quotient	
	Power of a Quotient	$\frac{6^3}{25^3}$
$\left(\frac{2}{5}\right)^{-2}$	Negative Exponents	
$11^3 \cdot 11^{-5}$	Product of Powers	
	Negative Exponents	$\frac{1}{45}$

3. Marques, Pablo, and Tandice used the product of powers law to write an equivalent expression for $5^{-7} \cdot 5^6$. Each student's equivalent expression is shown in the table.

Marques	Pablo	Tandice
$\frac{1}{5}$	-5	5^{-1}

   a. Which student's answer is not an equivalent expression for $5^{-7} \cdot 5^6$?

   b. Which student also applied the negative exponent law?

4. Find the value of the variable(s) that makes the equation true.

a. $7^{10} = 7^5 \cdot 7^f$

$f =$ ____

b. $\sqrt[3]{h} \cdot \sqrt[3]{h} \cdot \sqrt[3]{h} = -13$

$h =$ ____

c. $\sqrt{m} \cdot \sqrt{m} = 26$

$m =$ ____

d. $\frac{7^5}{6^5} = \left(\frac{6}{7}\right)^y$

$y =$ ____

e. $35^{12} = (35^x)^2$

$x =$ ____

f. $\left(\frac{9}{10}\right)^{-1} = \frac{c}{r}$

$c =$ ____

$r =$ ____

g. $18^0 = \frac{18^w}{18^7}$

$w =$ ____

1.4.5 Wrap-Up: Cool Down

1. Use the symbols <, >, or = to compare each pair of values. Then, explain your choice.

Expression	Explanation
2^3 ______ $2 \cdot 3$	
5^2 ______ 2^5	
100^1 ______ 1^{100}	

Unit 1, Lesson 5: Exploring Rational Exponents

Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Learning Targets

- ☐ I can apply the Laws of Exponents to rational exponents.
- ☐ I can convert between expressions involving rational exponents and expressions involving radicals.

1.5.1 Warm-Up: Bell Work

1. Use the Laws of Exponents to write an equivalent expression. Then, find the value of each expression. If necessary, use a calculator to find the value of the equivalent expression.

$(8^{-5})(8^{3})$ Equivalent Expression _________ Value __________________	$(3^{4})^{6}$ Equivalent Expression _________ Value __________________
$\left(\frac{2}{11}\right)^{3}$ Equivalent Expression _________ Value __________________	$\left(\frac{12}{36}\right)^{-4}$ Equivalent Expression _________ Value __________________

1.5.2 Exploration: Rational Exponents

1. Fill in the blank with the value of each.

$16^0 =$ ________ $16^1 =$ ________

2. Using the values above, guess the value of $16^{\frac{1}{2}}$.

3. Explain the strategy you used.

4. Find the value of $16^{\frac{1}{2}}$ using a calculator.

5. Discuss with your partner what you notice about the relationship between 16 and the value of $16^{\frac{1}{2}}$.

6. Predict the value of $49^{\frac{1}{2}}$. Check using your calculator.

7. Predict the value of $100^{\frac{1}{2}}$. Check using your calculator.

8. Write an expression equivalent to $a^{\frac{1}{2}}$.

9. Check your answer with your partner. If necessary, update your answer.

10. Predict the value of $8^{\frac{1}{3}}$. Check using your calculator.

11. Predict the value of $125^{\frac{1}{3}}$. Check using your calculator.

12. Write an expression equivalent to $a^{\frac{1}{3}}$.

13. Check your answer with your partner. If necessary, update your answer.

1.5.3 Guided Instruction: Why Does That Work?

1. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}}$	$6^{\_\_+\_\_} = 6^{\_\_}$	

2. What is the whole number that completes this equation?

$$\sqrt{6} \cdot \sqrt{6} = \_\_\_\_$$

Because $6^{\frac{1}{2}} \cdot 6^{\frac{1}{2}} = \_\_\_\_\_$ and $\sqrt{6} \cdot \sqrt{6} = \_\_\_\_\_$, then $6^{\frac{1}{2}}$ is equivalent to _______.

3. Use the product of powers law, $a^m \cdot a^n = a^{m+n}$, to evaluate the expression.

Expression	Product of Powers	Value
$5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}}$	$5^{\_\_+\_\_+\_\_} = 5^{\_\_}$	

4. What is the whole number that completes the equation?

$$\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = \_\_\_\_$$

Because $5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} \cdot 5^{\frac{1}{3}} = \_\_\_\_\_$ and $\sqrt[3]{5} \cdot \sqrt[3]{5} \cdot \sqrt[3]{5} = \_\_\_\_\_$,

then $5^{\frac{1}{3}}$ is equivalent to _______.

5. Rewrite each exponential expression as a radical expression. Then, evaluate each expression.

Exponential Expression	Radical Expression	Value
$81^{\frac{1}{2}}$		
$8^{\frac{1}{3}}$		
$(-216)^{\frac{1}{3}}$		

1.5.4 Exploration: Other Roots

Rewriting exponential expressions as radical expressions does not only apply to square roots and cube roots. Other roots apply these same principals.

Complete the table to explore other rational exponents and roots.

Rational Exponent	Radical Expression	Value
$81^{\frac{1}{4}}$		
	$\sqrt[4]{256}$	
	$\sqrt[5]{-32}$	
$\left(1{,}000{,}000{,}000^{\frac{1}{9}}\right)$		

1.5.5 Guided Instruction: Exponent Laws with Rational Exponents

1. Kalini's teacher asked her to rewrite $\left((100)^{\frac{1}{2}}\right)^3$ as an equivalent exponential expression. Kalini says that she can use a law of exponents to multiply the powers.

   a. Which law of exponents is Kalini using?

   b. Using the laws of exponents, what is an equivalent exponential expression for $\left((100)^{\frac{1}{2}}\right)^3$?

2. The expression $27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}}$ can also be rewritten using the laws of exponents.

   a. Rewrite the expression.

   $27^{\frac{1}{3}} \cdot 27^{\frac{5}{3}} =$

   b. Which exponent law was used?

   c. Check your answer with your partner.

   d. Evaluate the exponential expression.

3. Maeko explained his work for rewriting the following expression.

Explanation	$16^{-\frac{8}{2}} \cdot 16^{\frac{1}{2}}$
I first used the product of powers to rewrite the expression.	$16^{-\frac{8}{2}\cdot\frac{1}{2}} = 16^{-\frac{8}{4}}$
Next, I simplified the fractional exponent.	4^{-2}
Finally, I rewrote the expression using the negative exponent law.	$\frac{1}{16}$

Check Maeko's work and explain any errors he may have made.

1.5.6 Try It: Writing Rational Exponents

Rewrite each expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$5^{\frac{26}{7}} \cdot 5^{\frac{9}{7}}$		
$\dfrac{23^{\frac{8}{3}}}{23^{\frac{2}{3}}}$		
$125^{\frac{7}{15}} \cdot 125^{-\frac{2}{15}}$		
$169^{-\frac{5}{22}} \cdot 169^{\frac{25}{22}} \cdot 169^{-\frac{31}{22}}$		

1.5.7 Wrap-Up: Ticket Out the Door

1. Find the value of the variable.

   a. $\sqrt{14} = 14^{\frac{1}{k}}$

   $k =$ ____

   b. $50^{\frac{1}{3}} = \sqrt[n]{50}$

   $n =$ ____

2. Complete the table by writing an equivalent expression using the laws of exponents. Then, evaluate the expression.

Expression	Equivalent Expression	Value
$6^{-\frac{1}{6}} \cdot 6^{\frac{19}{6}}$		
$\dfrac{7^{\frac{1}{2}}}{7^{\frac{5}{2}}}$		

Unit 1, Lesson 6: Writing Equivalent Expressions Using Rational Exponents

Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Learning Targets

- ☐ I can apply the Laws of Exponents to rational exponents.
- ☐ I can convert between expressions involving rational exponents and expressions involving radicals.

1.6.1 Warm-Up: Identifying Equivalent Expressions

Three lists of numbers are shown.

List A	List B	List C
$8 \cdot 8 \cdot 8$	2^9	24
$4 \cdot 16 \cdot 2$	$6 \cdot 4$	$\frac{1}{32}$
$2 \cdot 16$	$\left(\frac{1}{2}\right)^5$	512
$\frac{1}{4} \cdot \frac{1}{4} \cdot \frac{1}{2}$	$2^2 \cdot 2^3$	128
$4 + 4 + 4 + 4 + 4 + 4 + 4$	$2 \cdot 4^3$	32

1. Choose an expression from List A and match it with an equivalent expression from List B and List C.

2. Explain to your partner how you know the expressions you matched are equivalent.

3. Listen to your partner explain the match they found. If you disagree, discuss your thinking and work to reach an agreement.

1.6.2 Guided Practice: Writing Equivalent Expressions

1. Use the power of a power law of exponents to write an equivalent exponential expression for the following two expressions.

   a. Expression 1: $\left(a^{\frac{1}{c}}\right)^b = a^{(\_\_\_\_ \cdot \_\_\_\_)} =$ ______

   b. Expression 2: $(a^b)^{\frac{1}{c}} = a^{(\_\_\_\_ \cdot \_\_\_\_)} =$ ______

   c. Explain why the expressions $\left(a^{\frac{1}{c}}\right)^b$ and $(a^b)^{\frac{1}{c}}$ are equivalent.

2. Two students evaluated an expression using two different methods.

Jerry	Antonio
$\left((81)^{\frac{1}{2}}\right)^3$	$\left((81)^{\frac{1}{2}}\right)^3$
$(\sqrt{81})^3$	$81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}} \cdot 81^{\frac{1}{2}}$
$(9)^3$	$\sqrt{81} \cdot \sqrt{81} \cdot \sqrt{81}$
729	$9 \cdot 9 \cdot 9$
	729

Explain the difference between the two methods.

1.6.3 Exploration: Matching Expressions

1. Use your calculator to help you match each expression in the first table with an equivalent expression in the second table. Record your answers in the table below.

Rational Exponent
$(64)^{\frac{3}{2}}$
$(64)^{-\frac{1}{2}}$
$(64)^{\frac{2}{3}}$
$(64)^{\frac{1}{4}}$
$(64)^{\frac{1}{2}}$
$(64)^{\frac{4}{3}}$

Radical Expression
$\sqrt[4]{64}$
$\sqrt[3]{64^2}$
$\left(\sqrt[3]{64}\right)^4$
$\sqrt{64}$
$\left(\sqrt{64}\right)^3$
$\frac{1}{\sqrt{64}}$

Rational Exponent	Radical Expression	Rational Exponent	Radical Expression
$(64)^{\frac{3}{2}}$		$(64)^{\frac{1}{4}}$	
$(64)^{-\frac{1}{2}}$		$(64)^{\frac{1}{2}}$	
$(64)^{\frac{2}{3}}$		$(64)^{\frac{4}{3}}$	

2. Discuss with your partner the relationship between the rational exponent and the radical expression.

3. Work with your partner to complete the following statement.

$$\left(a^{\frac{1}{c}}\right)^{b} = \left(a^{b}\right)^{\frac{1}{c}} = \left(\underline{\quad}\sqrt{a}\right)^{\underline{\quad}} = \underline{\quad}\sqrt{\left(a^{\underline{\quad}}\right)}$$

1.6.4 Activity: Card Sort

1. Work with your partner to sort the cards into sets of equivalent expressions.

$(18^2)^{\frac{1}{3}}$	$\left(21^{\frac{1}{4}}\right)^5$	$18^{\frac{3}{2}}$	$(21^5)^{\frac{1}{4}}$
$\left(\frac{1}{21^4}\right)^{\frac{1}{5}}$	$\frac{1}{(\sqrt[4]{21})^{-5}}$	$21^{\frac{5}{4}}$	$(18^3)^{\frac{1}{2}}$
$\left(18^{\frac{1}{2}}\right)^3$	$\left(21^{\frac{1}{5}}\right)^{-4}$	$18^{\frac{2}{3}}$	$\frac{1}{(\sqrt[5]{21})^4}$
$21^{-\frac{4}{5}}$	$(\sqrt[3]{18})^2$	$(\sqrt{18})^3$	$\left(18^{\frac{1}{3}}\right)^2$

Record your matches below.

Equivalent Expression

1.6.5 Try It: Writing Equivalent Expressions

1. Complete the table.

$(a)^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^{b}$	$(a^{b})^{\frac{1}{c}}$	$(\sqrt[c]{a})^{b}$	$\sqrt[c]{a^{b}}$
$(100)^{\frac{3}{2}}$	$\left((100)^{\frac{1}{2}}\right)^{3}$			
			$(\sqrt[4]{60})^{5}$	
				$\sqrt[5]{8^{3}}$
		$((27)^{2})^{\frac{1}{3}}$		

2. Based on the equivalent expressions in the table, write three equivalent expressions for $(\sqrt{17})^{8}$.

1.6.6 Your Turn!

1. Rewrite each expression using a rational exponent in the form $(a)^{\frac{b}{c}}$, where a, b, and c are integers.

$(\sqrt[3]{72})^{5}$

$\sqrt{5^{3}}$

$\dfrac{1}{(\sqrt[3]{20})^{4}}$

$\dfrac{1}{\sqrt{6}}$

1.6.7 Wrap-Up: Cool Down

1. Circle the expressions equivalent to $(25^3)^{\frac{1}{2}}$.

$(25)^{\frac{3}{2}}$ $\left(\sqrt{25}\right)^3$ $(25)^{\frac{2}{3}}$ $(25^{\frac{1}{3}})^2$ $\sqrt{25^3}$

Unit 1, Lesson 7: Evaluating Exponential Expressions

Benchmark

MA.912.NSO.1.1 Extend previous understanding of the Laws of Exponents to include rational exponents. Apply the Laws of Exponents to evaluate numerical expressions and generate equivalent numerical expressions involving rational exponents.

Learning Targets

- ☐ I can evaluate numerical expressions involving rational exponents.
- ☐ I can generate equivalent numerical expressions involving rational exponents.

1.7.1 Warm-Up: Perfect Squares Puzzle

1. Complete the table by finding the square of each whole number.

Whole Number	Square	Whole Number	Square	Whole Number	Square
1	$1^2 = 1$	8		15	
2	$2^2 = 4$	9		16	
3	$3^2 = 9$	10		17	
4		11		18	
5		12		19	
6		13		20	
7		14		21	

2. Place a digit in each blue box so that the principal square root is a whole number, and the expression is as close to 0 as possible. Use only the digits 1 to 9, at most one time each.
   Reminder: The principal square root is the positive square root.

$$\sqrt{\square\,\square\,\square} - \sqrt{\square\,\square} - \sqrt{\square}$$

1.7.2 Collaboration: Simplifying Expressions with Exponents

1. With a partner, write the expression $(27)^{\frac{2}{3}}$ in three equivalent forms.

$(27)^{\frac{2}{3}}$			

2. Explain which equivalent form makes it easiest to find the value of $(27)^{\frac{2}{3}}$ without a calculator.

3. Find the value of $(27)^{\frac{2}{3}}$ without a calculator.

4. Verify the value of $(27)^{\frac{2}{3}}$ using your calculator.

5. For each expression, write an equivalent radical expression. Then, evaluate the expression using your calculator.

Expression	Equivalent Radical Expression	Value of the Expression
$(144)^{-\frac{3}{2}}$		
$-(-64)^{\frac{1}{3}}$		
$(225)^{-\frac{1}{2}}$		
$-9^{\frac{5}{2}}$		
$\left(\frac{81}{256}\right)^{\frac{3}{4}}$		

1.7.3 Activity: Exponents Maze

1. Work with a partner to complete the maze. Beginning at the "start" box, connect each expression to an equivalent expression to form one continuous path to the "end" of the maze. Move only horizontally or vertically.

Start $8^{\frac{4}{3}}$	$\left(8^{\frac{1}{4}}\right)^3$	$64(2)^{-2}$	$48(3)^{-1}$	$48(9)^{-\frac{1}{2}}$	$48(27)^{-\frac{1}{3}}$	$\left(\frac{48}{3}\right)^{-1}$
$\left(8^{\frac{1}{3}}\right)^4$	$(8^3)^{\frac{1}{4}}$	$8(8)^{\frac{1}{3}}$	$(8)^2(3)^{-1}$	$(8)^{\frac{1}{3}}(36)^{\frac{1}{2}}$	$32(2)^{-1}$	$\left(\frac{2}{32}\right)^{-1}$
$(8^4)^{\frac{1}{3}}$	$\left(\sqrt{64}\right)^2$	$(64)^{\frac{1}{2}}(2)$	$(2)^3(4)^{\frac{1}{2}}$	$8(36)^{\frac{1}{2}}$	$32(4)^{\frac{1}{2}}$	$\left(\frac{1}{2^4}\right)^{-1}$
$(8^2)^{\frac{2}{3}}$	$(64^2)^{\frac{1}{3}}$	$(4)^{\frac{1}{2}}(4)^3$	$(4)^{\frac{9}{2}}(4)^{-\frac{5}{2}}$	$(4)^{\frac{1}{2}}(4)^{-\frac{3}{2}}$	$\left(\sqrt{4}\right)^4$	$\left(4^{\frac{1}{2}}\right)^4$
$(64)^{\frac{3}{2}}$	$64^{\frac{2}{3}}$	$(4)^{\frac{1}{3}}(4)^{\frac{1}{3}}$	$(4)^{\frac{3}{5}}(4)^{\frac{7}{5}}$	$(2)^{\frac{3}{4}}(2)^{\frac{5}{4}}$	$(2)^{-\frac{1}{2}}(2)^{\frac{9}{2}}$	$(8)^{-4}$
$\left(\sqrt{64}\right)^3$	$\left(64^{\frac{1}{3}}\right)^2$	$\left(\sqrt[3]{64}\right)^2$	$(7)^0(4)^2$	$(2)^7(4)^{-1}$	$(2)^6(2)^{-2}$	$(2)^5(2)^{-1}$
$(64^3)^{\frac{1}{2}}$	$(4)^{\frac{3}{2}}$	$(4^3)^{\frac{1}{2}}$	$(4)^3(2)^4$	$(8^7)^{-1}$	$(2)^{\frac{3}{4}}$	$\left(\sqrt[3]{8}\right)^4$ End

2. What is the value of the expression in each box that created the path through the maze?

1.7.4 Your Turn!

1. Evaluate each expression without a calculator.

$(100)^{\frac{1}{2}} \cdot (2^3)$	$(8^{-1})\left(4^{\frac{5}{2}}\right)$
$(81^6)^{\frac{1}{24}}$	$(125)^{\frac{2}{3}}$
$\left(49^{\frac{1}{2}}\right)(3^{-2})$	$-(3^{12})^{\frac{1}{4}}$
$(2^4)\left(8^{-\frac{2}{3}}\right)$	$\left(\frac{15}{45}\right)^{-4}$
$-\left(\frac{3}{48}\right)^{\frac{3}{2}}$	$\left(\frac{144}{169}\right)^{-\frac{1}{2}}$

1.7.5 Wrap-Up: Reflection

Complete the following statements about today's lesson.

1. Today I am still unsure about …

2. To fill in this gap I intend to …

Unit 1, Lesson 8: Equivalent Radical Expressions

Benchmark

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.

Learning Targets

- ☐ I can relate the power of a product law of exponents to the product law of radicals.
- ☐ Given a numerical radical, I can use the product law of radicals to write an equivalent numerical radical.

1.8.1 Warm-Up: Error Analysis

Leilana and Khamari wrote different numerical expressions and expanded them to show each has a value of 16.

Leilana's Work	Khamari's Work
$2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$	$2^8 = 2 \cdot 8 = 16$

1. Explain whether you agree with Leilana's work or Khamari's work.

2. Write another numerical expression involving exponents with a value of 16.

1.8.2 Exploration: Equivalent Radical Expressions

Ameer was asked to write an equivalent expression for $\sqrt{18}$. His work and resulting expression are shown.

Given Expression	Ameer's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (6 \cdot 3)^{\frac{1}{2}} = (6)^{\frac{1}{2}} \cdot (3)^{\frac{1}{2}}$	$\sqrt{6} \cdot \sqrt{3}$

Sophia wrote a different equivalent expression for $\sqrt{18}$. Her work and resulting expression are shown.

Given Expression	Sophia's Work	Equivalent Expression
$\sqrt{18}$	$18^{\frac{1}{2}} = (9 \cdot 2)^{\frac{1}{2}} = 9^{\frac{1}{2}} \cdot 2^{\frac{1}{2}} = \sqrt{9} \cdot \sqrt{2}$	$3\sqrt{2}$

1. What similarities and differences do you notice between Ameer's work and Sophia's work?

Similarities	Differences

2. With your partner, complete the work to find equivalent expressions for $\sqrt{175}$ two different ways.

Given Expression	Work	Equivalent Radical Expression
$\sqrt{175}$	$175^{\frac{1}{2}} = (35 \cdot 5)^{\frac{1}{2}} = (\_\_)^{\frac{1}{2}} \cdot (\_\_)^{\frac{1}{2}}$	$\sqrt{\_\_} \cdot \sqrt{\_\_}$
$\sqrt{175}$	$175^{\frac{1}{2}} = (25 \cdot 7)^{\frac{1}{2}} = (\_\_)^{\frac{1}{2}} \cdot (\_\_)^{\frac{1}{2}}$	$\sqrt{\_\_} \cdot \sqrt{\_\_}$

3. Which equivalent radical expression above can be rewritten in the form of an **integer** multiplied by a **radical**?

$\sqrt{\_\_\_} \cdot \sqrt{\_\_\_} = \_\_\_\sqrt{\_\_\_}$ because ______ is a perfect square, therefore $\sqrt{\_\_\_} = \_\_\_$.

1.8.3 Guided Instruction: Equivalent Radical Expressions

Recall the ***power of a product*** law of exponents. It states the product of two or more numbers raised to a power is equal to the product of each number raised to the same power.

$$(ab)^m = a^m \cdot b^m \quad \text{or} \quad a^m \cdot b^m = (ab)^m$$

The ***product law of radicals*** works similarly. Each of these equivalent expressions demonstrates the product law of radicals.

$\sqrt{18} = \sqrt{6} \cdot \sqrt{3}$
$\sqrt{18} = \sqrt{9} \cdot \sqrt{2}$

$\sqrt{175} = \sqrt{35} \cdot \sqrt{5}$
$\sqrt{175} = \sqrt{25} \cdot \sqrt{7}$

When a radical expression is rewritten as a product of its factors, each of the radicals will have the same index.

Product Law of Radicals

$\sqrt[m]{ab} = \sqrt[m]{a} \cdot \sqrt[m]{b}$ or $\sqrt[m]{a} \cdot \sqrt[m]{b} = \sqrt[m]{ab}$

1. Rewrite $\sqrt[3]{325}$ in two different ways using the product law of radicals.

When rewriting radical expressions, mathematicians generally try to rewrite the expression in the form of an **integer** multiplied by a **radical**, when possible.

$\sqrt{18} = \sqrt{9} \cdot \sqrt{2}$, which can be rewritten as $3\sqrt{2}$

$\sqrt{175} = \sqrt{25} \cdot \sqrt{7}$, which can be rewritten as $5\sqrt{7}$

2. What was done to rewrite the example expressions?

3. The following radical expressions are equivalent to $\sqrt[3]{320}$.

$\sqrt[3]{32} \cdot \sqrt[3]{10}$ $\quad$ $\sqrt[3]{64} \cdot \sqrt[3]{5}$

$\sqrt[3]{8} \cdot \sqrt[3]{40}$ $\quad$ $\sqrt[3]{16} \cdot \sqrt[3]{20}$

a. Determine which of the expressions can be rewritten in the form of an integer multiplied by a radical.

b. Discuss with your partner why these expressions can be rewritten.

4. Some radical expressions are given as an integer multiplied by the radical before any rewriting is done. Priya and Raj were asked to rewrite this type of expression. Their work is shown.

Priya's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 6\sqrt{5}$
Raj's Work	$3\sqrt{20} = 3 \cdot \sqrt{4} \cdot \sqrt{5} = 3 \cdot 2\sqrt{5} = 5\sqrt{5}$

Explain whether Priya or Raj rewrote the expression correctly.

1.8.4 Try It: Equivalent Radical Expressions

1. Write two equivalent expressions for each radical expression. Show all your steps.

a.

$\boldsymbol{\sqrt{135}}$
$\sqrt{135} =$
$\sqrt{135} =$

b.

$\boldsymbol{\sqrt[3]{600}}$
$\sqrt[3]{600} =$
$\sqrt[3]{600} =$

2. Write an equivalent expression for the following radical expression.

$$5\sqrt[3]{56}$$

1.8.5 Your Turn!

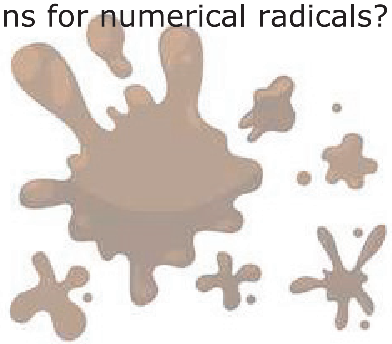
1. Select all expressions that are equivalent to $\sqrt{40}$.

- ☐ $(40)^{\frac{1}{2}}$
- ☐ $2\sqrt{20}$
- ☐ $\sqrt{4} \cdot \sqrt{10}$
- ☐ $(20 \cdot 2)^{\frac{1}{2}}$
- ☐ $5\sqrt{8}$
- ☐ $2\sqrt{10}$
- ☐ $(10 \cdot 4)^{\frac{1}{2}}$

2. Write an equivalent expression for $\sqrt[3]{378}$ in the form of a constant multiplied by a radical.

1.8.6 Wrap-Up: Reflection

1. What was the ***muddiest*** (most difficult or confusing) point in today's lesson on writing equivalent expressions for numerical radicals?



Unit 1, Lesson 9: Adding and Subtracting Radical Expressions

Benchmark

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.

Learning Target

- ☐ I can add or subtract two square roots or two cube roots.

1.9.1 Warm-Up: Adding and Subtracting Algebraic Expressions

1. Find the sum or difference of each algebraic expression.

   $4x + 23x$ $-7t - 40t$ $-35b + 10b$

2. Complete the statement to explain why is it possible to combine the terms in each of these expressions.

   When adding and subtracting algebraic expressions, __________ terms can be combined.

3. Which, if any, of the radical expressions do you think can be combined?

   - ☐ $\sqrt{17} + \sqrt{32}$
   - ☐ $5\sqrt{13} - 8\sqrt{3}$
   - ☐ $-7\sqrt{17} + 3\sqrt{17}$
   - ☐ $9\sqrt{5} + \sqrt{5}$
   - ☐ $8\sqrt{40} - 4\sqrt{9}$
   - ☐ $11\sqrt{48} - 6\sqrt{48}$
   - ☐ $-2\sqrt{49} - 3\sqrt{4}$

1.9.2 Exploration: Adding and Subtracting Radical Expressions

1. In small groups, determine the value of each expression.

$$\sqrt{25} + \sqrt{100} \qquad\qquad \sqrt{25+100}$$

2. Explain which of the expressions is equivalent to $\sqrt{125}$.

3. Paloma says $\sqrt{16} - \sqrt{9} = \sqrt{7}$. Ty'rique says $\sqrt{16} - \sqrt{9} = 1$. Who is correct? Verify by entering $\sqrt{16} - \sqrt{9}$ into a calculator.

   ____________ is correct because $\sqrt{16} - \sqrt{9}$ is equal to

   _____.

4. Xavier says $\sqrt{12} + 2\sqrt{12} = 3\sqrt{12}$. Sam says $\sqrt{12} + 2\sqrt{12} = 2\sqrt{24}$. Who is correct? Verify using a calculator.

   ____________ is correct because $\sqrt{12} + 2\sqrt{12}$ is equal to

   _______.

1.9.3 Guided Instruction: Adding and Subtracting Radical Expressions

When adding and subtracting algebraic expressions, only ***like terms*** can be combined. Adding and subtracting radical expressions is similar because only like radicals can be combined.

> **Like radicals** are radical terms that have the same index and the same radicand.

1. Find the sum or difference of each radical expression.

   a. $4\sqrt{5} + 7\sqrt{5} =$ ____________

   b. $30\sqrt[3]{2} - 23\sqrt[3]{2} =$ ____________

   c. $-10\sqrt{7} - 23\sqrt[3]{7} =$ ____________

   d. $-\sqrt{17} + \sqrt{17} =$ ____________

Three steps can be used when adding and subtracting radical expressions.

Step 1: Find perfect squares (for square roots) or perfect cubes (for cube roots) that are factors of the radicand.

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares or cubes.

Step 3: Add or subtract like radical terms.

2. Use the steps to add the following radical expression.

$$3\sqrt{6}+5\sqrt{24}$$

Step 1: Both terms include square roots. If possible, find perfect squares that are factors of the radicands, 6 and 24.

Step 2: Rewrite the radical expressions by applying the product law of radicals and evaluating any perfect squares.

$$3\sqrt{6}+5\sqrt{24}=\_\_\_\_\_\_\_\_\_\_+\_\_\_\_\_\_\_\_\_\_$$

Step 3: Add or subtract like radical terms.

3. Three students added $\sqrt{180}$ and $\sqrt{45}$ differently. Their work is shown.

Gerald's Work	Kalo's Work	Paola's Work
$\sqrt{180}+\sqrt{45}$	$\sqrt{180}+\sqrt{45}$	$\sqrt{180}+\sqrt{45}$
$2\sqrt{45}+\sqrt{45}$	$3\sqrt{20}+3\sqrt{5}$	$6\sqrt{5}+3\sqrt{5}$
$3\sqrt{45}$		$9\sqrt{5}$

a. Use a calculator to verify if the students' answers are equivalent.

b. Why did the students end up with different expressions for their sum?

c. Show how Gerald and Kalo could continue their work to arrive at the same expression as Paola.

Gerald's Next Steps	Kalo's Next Steps
$3\sqrt{45}$	$3\sqrt{20}+3\sqrt{5}$

4. Megh showed her work on the problem.

$$\sqrt[3]{32}+\sqrt[3]{64}=\sqrt[3]{8\cdot 4}+4=2\sqrt[3]{4}+4=6\sqrt[3]{4}$$

Write an explanation to Megh explaining the error in her work.

1.9.4 Guided Practice: Adding and Subtracting Radical Expressions

1. Select all of the expressions that can be combined. For the selected expressions, combine the terms. Show your work.

- ☐ $\sqrt{17}+\sqrt{32}$
- ☐ $5\sqrt{13}-8\sqrt{3}$
- ☐ $-7\sqrt{17}+3\sqrt{17}$
- ☐ $9\sqrt{5}+\sqrt{5}$
- ☐ $8\sqrt{40}-4\sqrt{9}$
- ☐ $11\sqrt{48}-6\sqrt{48}$
- ☐ $-2\sqrt{49}-3\sqrt{4}$

2. With a partner, find the sum and difference of the following problems. If you cannot add or subtract, circle the expression.

a. $\sqrt[3]{6}+3\sqrt{6}$

b. $\sqrt{20}-3\sqrt{45}$

c. $16^{\frac{1}{3}}-2^{\frac{1}{3}}$

d. $\sqrt{72}+\sqrt{18}-\sqrt{15}$

1.9.5 Your Turn!

1. Find the sum or difference. If you cannot add or subtract, circle the expression.

   a. $\sqrt[3]{27000} - \sqrt[3]{512}$

   b. $\sqrt{50} + \sqrt{18} + \sqrt{10}$

   c. $\sqrt[3]{6} - \sqrt[3]{48}$

   d. $32^{\frac{1}{2}} - 16^{\frac{1}{3}}$

1.9.6 Wrap-Up: Ticket Out the Door

1. Find the sum.

$$17\sqrt{8} + 5\sqrt{8}$$

Unit 1, Lesson 10: Multiplying and Dividing Radical Expressions

Benchmark

MA.912.NSO.1.4 Apply previous understanding of operations with rational numbers to add, subtract, multiply, and divide numerical radicals.

Learning Target

- [ ] I can multiply and divide two square roots or two cube roots.

1.10.1 Warm-Up: True Equations

1. Using the digits 0 to 9, place a digit in each blue box to make both equations true. Each digit can only be used one time, at most.

$$\sqrt{\square\,\square} = \square\,\sqrt{\square}$$

$$\sqrt{\square\,\square} = \square$$

1.10.2 Guided Instruction: Multiplying and Dividing Radical Expressions

Recall the ***quotient of powers*** law of exponents.

$$\frac{a^m}{a^n} = a^{m-n}$$

Ariana used the ***quotient of powers*** law of exponents to simplify $\frac{\sqrt{10}}{\sqrt{10}}$. Her work is shown.

$$\frac{\sqrt{10}}{\sqrt{10}} = \frac{10^{\frac{1}{2}}}{10^{\frac{1}{2}}} = 10^{\frac{1}{2}-\frac{1}{2}} = 10^0 = 1$$

When dividing radicals with the same index, the ***quotient law of radicals*** is used to rewrite the expression. Each of the following equivalent expressions demonstrates the quotient law of radicals.

$$\frac{\sqrt{13}}{\sqrt{13}} = \sqrt{\frac{13}{13}} = \sqrt{1} = 1 \qquad \frac{\sqrt{21}}{\sqrt{21}} = \sqrt{\frac{21}{21}} = \sqrt{1} = 1$$

1. Rewrite $\frac{\sqrt[3]{2}}{\sqrt[3]{2}}$ using the quotient law of radicals.

When dividing radical expressions, rewrite radical expressions in equivalent forms and use the quotient law to eliminate radicals where possible.

2. Write an equivalent expression for $\frac{\sqrt{125}}{\sqrt{245}}$.

$\frac{\sqrt{125}}{\sqrt{245}}$	
$\frac{\sqrt{125}}{\sqrt{245}} = \frac{\sqrt{\underline{\qquad} \cdot 5}}{\sqrt{\underline{\qquad} \cdot 5}}$	The radicands _______ and _______ are rewritten as the product of a factor and 5.
$\frac{\sqrt{\underline{\qquad} \cdot 5}}{\sqrt{\underline{\qquad} \cdot 5}} = \frac{\underline{\qquad} \cdot \sqrt{5}}{\underline{\qquad} \cdot \sqrt{5}}$	The radicals are rewritten using the ___________ law of radicals.
$\frac{\underline{\qquad} \cdot \sqrt{5}}{\underline{\qquad} \cdot \sqrt{5}} = \frac{\underline{\qquad}}{\underline{\qquad}} \cdot \frac{\sqrt{5}}{\sqrt{5}}$	Rewrite each perfect square root.
Because $\frac{\sqrt{5}}{\sqrt{5}} = 1$, $\frac{\underline{\qquad}}{\underline{\qquad}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\underline{\qquad}}{\underline{\qquad}}$.	

3. With your partner, rewrite the expression $\frac{\sqrt{275}}{\sqrt{44}}$ using the quotient laws of radicals.

4. Jean and Marcus were asked to explain how they would multiply $\sqrt{8} \cdot \sqrt{8}$. Their explanations are shown.

Jean's Explanation	I would first rewrite the radical expressions using rational exponents and then use the ***product of powers law of exponents*** to multiply the like bases. $\sqrt{8} \cdot \sqrt{8} = 8^{\frac{1}{2}} \cdot 8^{\frac{1}{2}} = 8^{\frac{1}{2}+\frac{1}{2}} = 8^{\frac{2}{2}} = 8^1 = 8$
Marcus's Explanation	I would use the ***product law of radicals*** to multiply the radical expressions and then use the law again to rewrite the square root. $\sqrt{8} \cdot \sqrt{8} = \sqrt{8 \cdot 8} = \sqrt{64} = 8$

Do you agree that both expressions have a value of 8?

5. Explain which method you would use to multiply $\sqrt{12} \cdot \sqrt{12}$.

6. Multiply the following radical expressions.

$$\sqrt{6} \cdot \sqrt{18}$$

In cases where there is a fraction with a radical in the **denominator**, mathematicians use a technique called **rationalizing the denominator** to rewrite an expression.

A model of how to rationalize the expression $\frac{30}{\sqrt{2}}$ is shown.

$$\frac{30}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{30\sqrt{2}}{\sqrt{4}} = \frac{30\sqrt{2}}{2} = 15\sqrt{2}$$

1.10.3 Activity: Pass Around

In groups of four, each student completes the **first** step of their problem. The students then pass their problem to the student on their right to complete the next step. For each new step, continue passing the problems to the student on the right.

Student A	**Student B**
$\dfrac{\sqrt[3]{200}}{\sqrt[3]{625}}$	$\sqrt[3]{14} \cdot \sqrt[3]{20}$
Student C	**Student D**
$\sqrt{24} \cdot \sqrt{30}$	$\dfrac{\sqrt{256}}{\sqrt{108}}$

1.10.4 Your Turn!

1. Write an equivalent expression for each radical expression.

   a. $\sqrt{9} \cdot \sqrt{9}$ b. $\frac{\sqrt[3]{56}}{\sqrt[3]{77}}$ c. $\frac{2\sqrt{22}}{\sqrt{8}}$

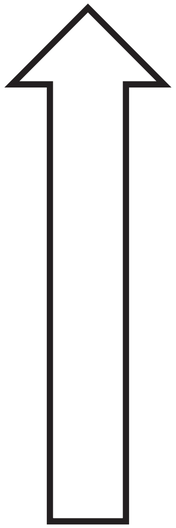
2. Anjelica was asked to rewrite the expression $\frac{\sqrt[3]{264}}{\sqrt[3]{81}}$. Anjelica's work is shown. Write an explanation to Anjelica describing why her problem is incorrect. Then, show how to correctly work the problem.

$$\frac{\sqrt[3]{264}}{\sqrt[3]{81}} = \frac{\sqrt[3]{8 \cdot 33}}{\sqrt[3]{9 \cdot 9}} = \frac{2\sqrt[3]{33}}{9}$$

1.10.5 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding of the following.

   I can multiply and divide two square roots or two cube roots.



I am so confident; I could explain this to someone else!

I can get to the right answer, but I do not understand well enough to explain it yet.

I understand some of this, but I do not understand all of it yet.

I tried hard and I listened, but I am finding this challenging. I will get help by …

I do not understand any of this yet. I will get help by ….

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